



# Nikoo Asadnejad

## Software Engineer

**Senior Software Engineer** with 5 years of experience designing and developing scalable, high-performance applications using **C#, .NET**, and modern software architecture principles in **Agile/Scrum** environment. Experienced in building **distributed systems** and business-critical platforms using **Microservices, Domain-Driven Design (DDD), CQRS, gRPC, RabbitMQ, Redis, MongoDB, SQL Server**, and **REST APIs**.

Strong background in **System Design, Software Architecture, Performance Optimization**, and **Observability**. Skilled in designing reliable, maintainable, and scalable solutions that align with business objectives and engineering best practices.

In addition to my core expertise in the Microsoft ecosystem, I have foundational knowledge of **Python** and **Go**, which I use to expand my capabilities in cloud-native development, and AI-related projects.

Passionate about Artificial Intelligence, **AI Engineering**, and **Agentic AI** systems. I regularly leverage AI-powered development tools such as **Claude Code** and **Cursor** to accelerate software delivery, improve developer productivity, and explore innovative engineering solutions.

Currently pursuing a **Master's** degree in Software Engineering, with research focused on AI-driven autoscaling for Kubernetes and cloud-native environments.

I enjoy collaborating with high-performing engineering teams, **solving complex technical challenges**, continuously learning new technologies, and building scalable products that deliver meaningful business impact.

## Experience

### Oct 2022 - Present | Fly Today

#### Senior Back-End Engineer

##### Responsibilities and Achievements:

- Took end-to-end ownership of *Train, Bus, Activities Booking* systems and *Hotel Recommendation* systems, from **requirements analysis** and **architecture design** through development, deployment, monitoring, and maintenance.
- Designed and developed scalable systems using **Clean Architecture, DDD, CQRS, Microservices, SOA**, and **Modular Monoliths**, with **gRPC, RabbitMQ, SignalR, YARP**, and **Ocelot**.
- Designed a Financial **Ledger** system using **Event Sourcing**, focusing on transaction traceability, consistency, and auditability.
- **Improved API throughput** and **performance** through **Redis** and in-memory **caching, async/parallel programming, SQL query optimization, MongoDB** indexing.
- Implemented an external **Wallet** system using **Chain of Responsibility, Strategy**, and **Factory** design patterns, focusing on extensibility and maintainability.
- Implemented *fuzzy* and *semantic* hotel search using **Elasticsearch** to improve search relevance.
- Improved observability and reliability using **OpenTelemetry, ELK, distributed tracing, metrics, health checks, Polly, retry policies**, and **distributed locking**.
- Designed **CI/CD** pipelines, containerized applications with **Docker**, and developed **unit and architectural tests** to ensure code quality and architectural consistency.
- Conducted **code reviews, mentored** junior developers, and **onboarding** new team members.

### Aug 2021 - Oct 2022 | Kahkeshan Group of Companies | Karan

#### Back-End Engineer

##### Responsibilities and Achievements:

- Developed new features and refactored legacy APIs in the Karan project
- (**12 microservices**), using ASP.NET Core with **SQL Server, Redis**, and **MongoDB**.
- Deployed and maintained a web crawler application and Telegram bots on **Linux** servers.
- **Optimized** search queries, achieving **3x faster** performance.
- Designed and implemented a multi-gateway payment service using the **Factory Pattern** for flexible payment integrations.
- Built a JobAlert matching service with **RabbitMQ** and **Quartz jobs** to automatically pair resumes with relevant job ads based on skills and requirements.

### Sep 2018 - Oct 2022 | Kahkeshan Group of Companies | Karan

#### Software Support and QA Specialist

## Education

2024-now

**Master / Software Engineering** | Azad University, Science And Research Branch  
2018-2022

**Bachelor / Software Engineering** | Azad University Tehran North Branch

## Contact

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### LinkedIn

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### GitHub

<https://github.com/Nikoo-Asadnejad>

### Medium

<https://medium.com/@nikoo.asadnejad.work>

## Technical Skills

- Software Engineering
- Software Architecture
- System Design
- MicroServices, Distributed Systems
- Clean Architecture, DDD, CQRS, Event-Sourcing
- Event-Driven, RabbitMQ
- Design Patterns, SOLID, OOP
- C#.Net, SQL Server, EF
- MongoDB, Redis, Elastic Search
- Restful Api, gRpc, SignalR, MVC, WCF, GraphQL
- GIT, CI/CD
- Unit Testing, xUnit, TDD
- Linux, Docker & Kubernetes
- Python & Bash Scripting
- HTML & CSS
- JavaScript & JQuery

## Language

English Professional

French Basic

# Licenses & Certifications

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- **Docker And Kubernetes**

CanDo Docker , Docker Swarm , Kubernetes  
Issued Feb 2025 60h

- **LIPC-2** 50h

Arjang Linux Kernel , Systemd , NGINX ,  
Issued Jul 2024

- **LIPC-1** 45h

CanDo | Linux Commands, Bash Scripts  
Issued Apr 2024

- **Asp.Net Core**

Kahkeshan Noor 20h 140h  
Issued Nov 2021 | .NET Core , Rest API

- **MCSA Web Applications**

Kahkeshan Noor  
Issued Aug 2020 | C# , .NET , ASP.NET , MVC

- **Web Design**

Tehran Institute Of Technology 80h  
Issued Apr 2019 | HTML , CSS , JavaScript , JQuery

## Projects

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- **NotificationService**

<https://github.com/Nikoo-Asadnejad/NotificationService>

A microservice-based notification platform built with asynchronous communication using RabbitMQ. It provides a central API for processing notification requests and dispatching them to dedicated Email, SMS, and Push Notification microservices, enabling scalable, reliable, and decoupled message delivery.

- **Linkedin Post Creator Agent**

[https://github.com/Nikoo-Asadnejad/post\\_creator\\_agent](https://github.com/Nikoo-Asadnejad/post_creator_agent)

AI LinkedIn Content Generator – A multi-agent AI system built with Python, LangChain (LCEL), and Ollama that automatically generates ready-to-publish LinkedIn posts and accompanying images. The workflow uses asynchronous parallel execution, where a search agent researches topics when needed, while content and image generation agents run concurrently to produce accurate, fact-based content with minimal hallucinations.

- **Kibana Dashboard Generator**

[https://github.com/Nikoo-Asadnejad/kibana\\_dashboard\\_generator](https://github.com/Nikoo-Asadnejad/kibana_dashboard_generator)

This project provides a fully automated solution for generating Kibana dashboards directly from a Swagger/OpenAPI specification. The script analyzes each controller (API scope) defined in the specification and produces a dedicated dashboard containing detailed monitoring panels for every endpoint. Each panel visualizes the API's operational metrics, including the distribution of returned HTTP status codes, enabling teams to observe performance patterns, detect anomalies, and quickly identify reliability issues.